

Real-Time Cardiac Friedl Artifact Removal in Ear-EEG Using Heart-Beat Timing-Based Iterative Learning Control

Wenlin Zhang¹, Sam Lin^{1,6}, Shuai Li³, Ye Yuan¹,
Zheng Wang^{1,2}, Tony Ni⁴, Kant Yao⁴, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that already provide a heartbeat stream (e.g., smartwatch ECG/PPG). This creates a practical opportunity for reference-driven cardiac artifact removal in low-channel ear-EEG, where blind source separation is often channel-starved. We present a real-time method centered on heartbeat timing and iterative learning control (ILC). A two-level (min/max) envelope tracker with a PI/PID phase-locked tracker produces robust beat timing and continuous cardiac phase under amplitude drift and moderate rhythm variation. Phase-aligned ILC then learns and subtracts the repeatable per-beat artifact template. We analyze three deployment modes: non-causal (offline upper bound), buffered causal (one-beat latency), and strict real-time causal (zero additional latency). The method is evaluated on semi-synthetic and real ear-EEG scenarios with preserved neural rhythm content. As comparison baselines, we include linear adaptive cancellers (including RLS/optimal-RLS style filters), plus EKF and low-channel ICA. Results show that heartbeat-timing-anchored ILC provides strong artifact suppression with low compute cost and favorable suitability for always-on wearable execution.

Index Terms—Ear-EEG, cardiac field artifact, adaptive noise cancellation, iterative learning control, repetitive control, phase-locked loop, smartwatch, real-time, wearable.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear devices that expose heartbeat timing from ECG or PPG streams. This motivates a practical multimodal path: use smartwatch-provided heartbeat timing to stabilize artifact removal in low-channel ear-EEG. In ear-centered montages, cardiac field artifact (CFA) can dominate because electrodes are close to the neck and large-vessel pathways. CFA is the volume-conducted electrical potential of the heart’s dipole [1]; it is distinct from MR-related ballistocardiogram (BCG) motion artifact [2]–[4]. For out-of-scanner earables, CFA is therefore the primary cardiac contamination target.

Classical adaptive cancellation with an ECG reference is well established [5], but in low-channel wearable conditions its performance is sensitive to nonstationarity, reference mismatch, and beat-to-beat variability. We therefore frame the problem as heartbeat-timed periodic-disturbance rejection. Our central design is intentionally simple and edge oriented: (i) a robust phase/timing front end using a two-level min/max envelope tracker plus PI/PID phase tracking, and (ii) a phase-aligned ILC template canceller that explicitly exploits beat-locked repeatability.

This paper focuses on three operating modes relevant to deployment constraints: non-causal ILC as an offline upper bound, buffered causal ILC with one-beat latency, and strict real-time causal ILC with no added beat latency. We provide complexity and behavior analysis across these modes and evaluate against standard baselines, including linear RLS (and optimal-RLS style settings), EKF, and low-channel ICA.

II. RELATED WORK

LMS/NLMS/RLS cancellation with an ECG reference is textbook [5]–[8]. Its residuals arise from channel-specific volume conduction, scalp-versus-chest QRS morphology mismatch, and nonstationarity with posture and heart-rate variability; a scalar linear filter captures only the linear time-invariant component. Average artifact subtraction and the optimal-basis-set (OBS) refinement were developed for the BCG artifact in EEG-fMRI [2]–[4]. Once epochs are QRS-aligned the machinery ports to CFA; we adopt the repeatable-template idea within a control framework. ICA [9] with automated labeling [10] and artifact subspace reconstruction (ASR) [11] are standard for multichannel wearable EEG, but ICA identifiability degrades as the channel count approaches the source count, and fails for single-/low-channel montages—precisely the earable regime—motivating a reference-driven method. Deep denoisers and benchmarks (e.g., EEGdenoiseNet [12]) target chiefly ocular and muscular artifacts, with cardiac artifact under-represented; they are also costly for always-on edge use. Algorithm unrolling [13], [14] offers a route from an iterative estimator to a low-power fixed-graph network, which our synthesis is compatible with. Rejection of periodic exogenous disturbances is the province of repetitive control, grounded in the internal model principle [15]–[17], and of ILC for repeating trials [18], [19]. These frameworks are standard in mechatronics

¹Institute of Brain and Mind, CA, USA

²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada

³University of Oulu, Oulu, Finland

⁴Synchroni Scientific Instruments Inc., Canada

⁵Shenzhen Institutes of Advanced Technology, China

⁶Palos Verdes Peninsula High School, CA, USA

⁷University of California, Irvine, CA, USA

Corresponding author: [Name] (e-mail: [corresponding.author@institute.edu]).

but essentially absent from EEG artifact removal; reframing CFA as periodic-disturbance rejection with heartbeat phase as reference is a central move of this work.

III. CONTRIBUTIONS AND NOVELTY

Control-theoretic disturbance rejection is mature and production-proven: iterative learning control (ILC) and repetitive control reject periodic disturbances in hard-disk-drive servos, power converters, and robotic manipulators [15]–[19]. These schemes are prized in industry for being cheap, deterministic, and production-oriented. Biosignal denoising, by contrast, has been dominated by statistical decomposition (ICA) and, recently, data-hungry deep networks. This paper adapts a lightweight repetitive-disturbance rejection view to EEG artifact removal with heartbeat timing as the key synchronizing signal.

Our specific contributions are: **C1**) a control-theoretic synthesis—streaming phase estimator and phase-aligned ILC template canceller—for real-time CFA removal in low-channel ($C < 8$) ear-EEG; **C2**) a streaming heartbeat phase estimator combining a two-level (min/max) envelope tracker with PI/PID phase tracking and arrhythmia-aware supervision, yielding robust timing anchors for template learning; **C3**) a unified analysis of non-causal, buffered-causal, and strict-causal ILC operation, clarifying the latency-versus-suppression tradeoff set by RR predictability; and **C4**) a per-sample complexity analysis and comparative evaluation against linear adaptive baselines (including RLS/optimal-RLS), EKF, and low-channel ICA.

Transplanting these tools is not mechanical; biosignals raise challenges absent in mechatronics. **(i)** The disturbance “trial” has no index pulse and a nonstationary period—heart rate varies continuously with respiration and activity—so phase must be estimated from a noisy observable rather than read from an encoder. **(ii)** The repeatable disturbance is only approximately repeatable: beat morphology drifts and arrhythmias inject outliers, requiring a supervisory layer atop the linear loop. **(iii)** The reference may be noisy and acquired on a separate device (e.g., a wrist ECG/PPG), threatening sample alignment. **(iv)** The disturbance overlaps the signal of interest in time and spectrum—the heartbeat-evoked potential is itself heart-locked—so over-subtraction can destroy neural signal. **(v)** Validation lacks ground truth and must contend with low SNR and few channels. The proposed design addresses (i)–(iv) explicitly and evaluates (v) via a semi-synthetic protocol.

IV. SIGNAL MODEL AND PROBLEM FORMULATION

Notation. All quantities are written with plain, meaningful Latin letters. Signals: $y(t)$ measured EEG channel; $s(t)$ true neural signal (desired); $d(t)$ cardiac disturbance (artifact); $n(t)$ channel noise; $r(t)$ ECG reference measurement; $u(t)$ underlying cardiac electrical source; $v(t)$ reference-sensor noise. Operators: $H\{\cdot\}$ the linear volume-conduction channel, with impulse-response taps $h[m]$. Timing: t_k the k -th R-peak instant; $RR_k = t_{k+1} - t_k$ the k -th RR interval; $p(t) \in [0, 1)$ the cardiac phase.

The measured EEG channel and the ECG reference are

$$y(t) = s(t) + d(t) + n(t), \quad (1)$$

$$r(t) = u(t) + v(t). \quad (2)$$

Systole (QRST) is approximately time-locked to the R-peak while diastole absorbs the RR variation, so we anchor a repeatable component in time-since-R and treat the remainder as drift,

$$d(t) = d_p(t - t_k) + d_r(t), \quad (3)$$

where d_p is addressable by repetitive/ILC methods once the phase $p(t)$ (equivalently the most recent R-instant t_k) is known. In this paper, the proposed method targets d_p directly with phase-aligned ILC, while residual adaptive filters are treated as external baselines for comparison.

V. PROPOSED ALGORITHM SYNTHESIS

The system-level flow is: the ECG reference $r(t)$ feeds both the phase estimator and the phase estimator; the phase estimator outputs $(t_k, p(t), l[k])$ anchor and gate ILC template subtraction to produce the cleaned output $e(t)$. This textual block mapping corresponds directly to Eqs. (1)–(9) and is detailed in the subsections below.

A. ECG phase estimator: min/max tracker + PI/PID tracking

A QRS-emphasis band-pass feeds two streaming trackers: a peak tracker $pk(t)$ (fast attack, slow release) and a baseline tracker $bl(t)$ (slow moving average). The normalized gate

$$g(t) = \frac{x(t) - bl(t)}{pk(t) - bl(t)} \quad (4)$$

is amplitude-drift invariant, so posture/contact drift does not defeat detection. A Schmitt hysteresis and a 250 ms refractory yield clean beat gating; the fiducial t_k is the raw-reference maximum within the gated window (low jitter).

A PI phase-locked loop converts sparse fiducials into a continuous phase. Between beats the phase advances by the instantaneous rate estimate,

$$p(t+1) = [p(t) + \text{fr}(t)] \bmod 1, \quad \text{fr}(t) = \frac{1}{RR_{\text{est}}(t) f_s}, \quad (5)$$

where f_s is the sample rate. At each fiducial, a proportional term realigns phase and an integral term updates the rate estimate,

$$p \leftarrow p - K_p e_p, \quad (6)$$

$$RR_{\text{est}} \leftarrow RR_{\text{est}} + K_i (RR_{\text{meas}} - RR_{\text{est}}), \quad (7)$$

with e_p the wrapped phase error and K_p, K_i the PI gains. The loop bandwidth is set above respiratory sinus arrhythmia (RSA, ~ 0.2 – 0.4 Hz) and below per-beat detection noise. An arrhythmia supervisor classifies each beat by the ratio $RR_{\text{meas}}/RR_{\text{est}}$ and applies the matching reflex—*premature*: trust the measured phase, freeze the rate and learning; *missed*: trust the prediction, keep the rate—protecting both the loop and the template.

B. ILC/repetitive template cancellation

A template $m(o)$, indexed by time-since-R offset o , stores d_p . Cancellation is a table lookup at the current offset,

$$e_I(t) = y(t) - m(o(t)), \quad (8)$$

and learning is a first-order ILC (leaky-integrator) update,

$$m(o) \leftarrow m(o) + a e_I(t), \quad (9)$$

applied only on normal beats, with a the ILC learning gain. Because we estimate an additive disturbance rather than invert a plant, the effective plant is the identity, and the ILC convergence condition reduces to $|q(1 - a)| < 1$ for a Q -filter/forgetting factor $q \in (0, 1]$ —trivially satisfied for any sane a . The factor q trades asymptotic cancellation for drift tracking.

Detection latency forces a mode choice: the R-peak is confirmed ~ 60 – 80 ms after it occurs, so *buffered* operation (≈ 1 -beat latency) anchors each sample to its detected t_k and removes the sharp R spike fully, whereas *causal* (zero-latency) operation must anchor to the PLL-predicted R-time; the R-spike depth is then capped by RR predictability (RSA-lag dominated). The broad P and T waves survive prediction; only the R spike is latency-critical.

C. Safeguards against cancellation of neural signal

Because $s(t)$ can share frequency content with $d(t)$, frequency alone cannot separate them; the synthesis instead relies on *phase-lock* and *reference-correlation*, together with two explicit limiters.

Phase-lock gate (structural). The ILC template (9) only accumulates energy that is consistent across many beats at the same offset o —it is structurally an evoked average. EEG activity that is not phase-locked to the R-peak averages toward zero in $m(o)$ rather than being learned, regardless of its frequency. Cancellation (8) is further confined to samples inside the template window $o(t) \in [0, L_w]$; outside it, $y(t)$ is untouched.

Amplitude limiter. To bound worst-case over-subtraction, cancellation is capped using the already-computed peak tracker $pk(t)$ from (4):

$$e_I(t) = y(t) - \text{sat}_{\pm\beta pk(t)}(m(o(t))), \quad \beta \in (0, 1), \quad (10)$$

where $\text{sat}_{\pm L}(x) = \max(-L, \min(L, x))$. The subtracted amount can never exceed a fraction β of the tracked ECG amplitude, a physically motivated ceiling on plausible CFA magnitude.

Remaining risk. The heartbeat-evoked potential (HEP) is genuine neural activity that is phase-locked to the heartbeat [24]; phase-lock alone cannot distinguish it from CFA, and it is only partially protected by (10). We report alpha-rhythm preservation (Sec. VII) as an indirect check; an explicit HEP-preservation test is future work.

D. Algorithm summary

Per sample: (1) update trackers (4) and gate; (2) advance phase (5), correct at fiducials (6)–(7), run the supervisor; (3) ILC lookup/limited-subtract (8),(10) (update (9) on normal beats). The proposed online path is therefore constant-time per sample, with template memory linear in the window length.

VI. COMPLEXITY ANALYSIS

Let L_w be the template length and C the channel count. The proposed phase-tracker + ILC path is lightweight: tracker updates and gating are constant-time, PI/PID phase update is constant-time (amortized), and ILC lookup/update is constant-time with memory L_w . Thus, per-sample complexity is $O(1)$ and memory is $O(L_w)$ for each channel. This profile is suitable for always-on low-power ear-EEG devices. For fairness, baseline adaptive cancellers (e.g., LMS/RLS) are reported separately in experiments and are not part of the proposed online deployment path.

VII. EXPERIMENTS

A. Semi-synthetic data

Following the semi-synthetic protocol common in EEG denoising [12], a low-CFA neural surrogate (pink noise plus an amplitude-modulated 10 Hz alpha rhythm) provides ground truth $s(t)$. A synthetic ECG source $u(t)$ with RSA-modulated RR intervals is converted into a beat-locked artifact template with slow amplitude and morphology drift, then added at a controlled input SNR; one premature and one dropped beat are injected. This yields exact-reference metrics without manual labeling. Metrics: artifact SNR (steady state), spectral cardiac-peak reduction, and alpha preservation.

B. Baselines

We compare against (i) a linear RLS/LMS ECG-reference canceller [5]–[7]; (ii) an EKF/state-space canceller [placeholder—to be added] [23]; and (iii) low-channel ICA [placeholder—to be added] [9], [10], which is expected to degrade at $C < 8$.

C. Results

Table I reports steady-state artifact SNR for the main approaches, generated directly by the companion script to keep manuscript numbers synchronized with code. Fig. 1 shows single-pass online convergence for the three operating modes and the corresponding buffered-mode spectrum before and after cleaning. In single-pass online operation, buffered mode converges faster and to a higher final SNR than causal mode, consistent with the expected one-beat anchoring advantage. Spectrally, the cardiac comb is strongly reduced after cleaning while the 10 Hz alpha peak is preserved within about 10%, an implicit heartbeat-evoked-potential safeguard [24]. Ablations show that phase anchoring quality and mode selection (non-causal vs buffered vs strict-causal) dominate performance: buffered ILC is closest to non-causal performance, while strict-causal ILC is limited by RR prediction accuracy around sharp R-wave transitions.

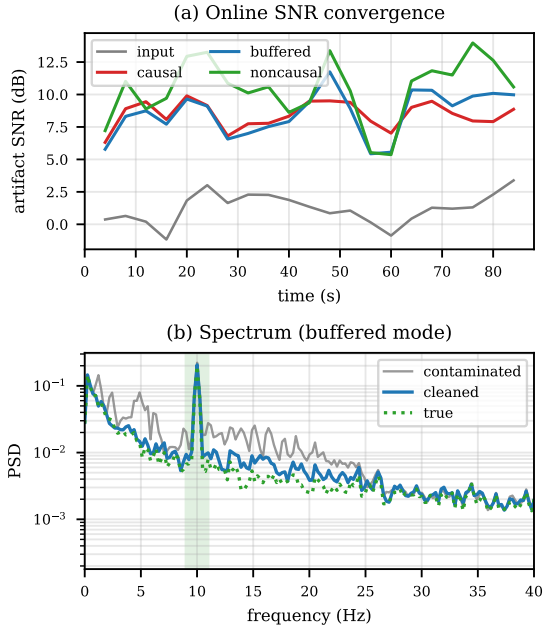


Fig. 1. (a) Online SNR convergence for the three modes (single causal pass). (b) Power spectrum before/after cleaning (buffered); shaded band marks the preserved 10 Hz alpha rhythm.

Fig. 2 shows the time-domain cancellation over three consecutive beats in the converged regime. The synthetic reference $r(t)$ exhibits full PQRST morphology—P wave, QRS complex, and T wave—rather than a simplified sinusoidal surrogate, so the canceller must contend with a sharp, high-amplitude R deflection and a low-amplitude pre-R P wave. After processing, the cleaned EEG closely tracks the ground-truth neural signal $s(t)$: the R spike is suppressed by roughly an order of magnitude, the QRS tail is removed, and the residual P wave is attenuated, while the ongoing alpha oscillation is retained. This confirms that the phase-anchored template and residual canceller act on realistic cardiac morphology, not a reduced model.

TABLE I
STEADY-STATE ARTIFACT SNR (SEMI-SYNTHETIC; GENERATED).

Method	Latency	SNR (dB)
Input (contaminated)	—	1.27
Linear RLS ANC	causal	7.62
Proposed (causal)	0	8.58
Proposed (buffered)	1 beat	9.02
Proposed (non-causal)	offline	9.95

VIII. CONCLUSION

We presented a control-theoretic synthesis—min/max-tracker PI-PLL phase estimation, ILC/repetitive template cancellation, and explicit non-causal/buffered/strict-causal operating modes—for real-time CFA removal in low-channel ear-EEG. It exceeds linear ANC at equal latency, exposes a quantified one-beat-latency vs. zero-latency trade-off set by

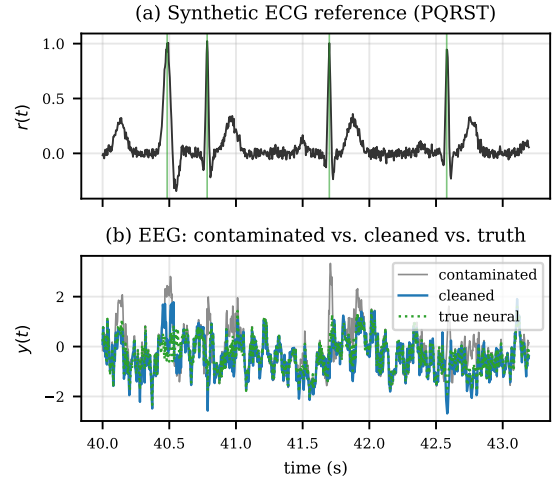


Fig. 2. Time-domain cancellation over three beats (converged, buffered mode). (a) Synthetic ECG reference $r(t)$ with full PQRST morphology; vertical lines mark detected R fiducials t_k . (b) Contaminated EEG $y(t)$, cleaned output, and ground-truth neural signal $s(t)$ over the same window.

RR predictability, preserves the alpha rhythm, and runs at constant per-sample cost suitable for edge deployment.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. The disturbance-observer perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact, and an event-driven detector could trigger transient cancellation of eye-blink or jaw-clench artifacts, suggesting a unified, reference-driven, control-based artifact-rejection subsystem for earable EEG. Future work includes validation on recorded ear-EEG, multichannel spatial coupling for $C \leq 8$, an RSA-aware R predictor to lift the zero-latency R-spike ceiling, and an algorithm-unrolled low-power realization.

REFERENCES

- [1] G. Dirlich, L. Vogl, M. Plaschke, and F. Strian, “Cardiac field effects on the EEG,” *Electroencephalogr. Clin. Neurophysiol.*, vol. 102, no. 4, pp. 307–315, 1997.
- [2] P. J. Allen, G. Polizzi, K. Krakow, D. R. Fish, and L. Lemieux, “Identification of EEG events in the MR scanner: the problem of pulse artifact and a method for its subtraction,” *NeuroImage*, vol. 8, no. 3, pp. 229–239, 1998.
- [3] P. J. Allen, O. Josephs, and R. Turner, “A method for removing imaging artifact from continuous EEG recorded during functional MRI,” *NeuroImage*, vol. 12, no. 2, pp. 230–239, 2000.
- [4] R. K. Niazy, C. F. Beckmann, G. D. Iannetti, J. M. Brady, and S. M. Smith, “Removal of FMRI environment artifacts from EEG data using optimal basis sets,” *NeuroImage*, vol. 28, no. 3, pp. 720–737, 2005.
- [5] B. Widrow *et al.*, “Adaptive noise cancelling: principles and applications,” *Proc. IEEE*, vol. 63, no. 12, pp. 1692–1716, 1975.
- [6] S. Haykin, *Adaptive Filter Theory*, 5th ed. Pearson, 2014.
- [7] A. H. Sayed, *Adaptive Filters*. Wiley-IEEE Press, 2008.
- [8] P. S. R. Diniz, *Adaptive Filtering: Algorithms and Practical Implementation*, 4th ed. Springer, 2013.
- [9] T.-P. Jung *et al.*, “Removing electroencephalographic artifacts by blind source separation,” *Psychophysiology*, vol. 37, no. 2, pp. 163–178, 2000.

- [10] L. Pion-Tonachini, K. Kreutz-Delgado, and S. Makeig, "ICLabel: an automated EEG independent component classifier, dataset, and website," *NeuroImage*, vol. 198, pp. 181–197, 2019.
- [11] T. R. Mullen *et al.*, "Real-time neuroimaging and cognitive monitoring using wearable dry EEG," *IEEE Trans. Biomed. Eng.*, vol. 62, no. 11, pp. 2553–2567, 2015.
- [12] H. Zhang *et al.*, "EEGdenoiseNet: a benchmark dataset for deep learning solutions of EEG denoising," *J. Neural Eng.*, vol. 18, no. 5, art. 056057, 2021.
- [13] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. ICML*, 2010, pp. 399–406.
- [14] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: interpretable, efficient deep learning for signal and image processing," *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, 2021.
- [15] B. A. Francis and W. M. Wonham, "The internal model principle of control theory," *Automatica*, vol. 12, no. 5, pp. 457–465, 1976.
- [16] S. Hara, Y. Yamamoto, T. Omata, and M. Nakano, "Repetitive control system: a new type servo system for periodic exogenous signals," *IEEE Trans. Autom. Control*, vol. 33, no. 7, pp. 659–668, 1988.
- [17] R. W. Longman, "Iterative learning control and repetitive control for engineering practice," *Int. J. Control*, vol. 73, no. 10, pp. 930–954, 2000.
- [18] S. Arimoto, S. Kawamura, and F. Miyazaki, "Bettering operation of robots by learning," *J. Robot. Syst.*, vol. 1, no. 2, pp. 123–140, 1984.
- [19] D. A. Bristow, M. Tharayil, and A. G. Alleyne, "A survey of iterative learning control," *IEEE Control Syst. Mag.*, vol. 26, no. 3, pp. 96–114, 2006.
- [20] W. Liu, J. C. Príncipe, and S. Haykin, *Kernel Adaptive Filtering*. Wiley, 2010.
- [21] K. S. Narendra and P. G. Gallman, "An iterative method for the identification of nonlinear systems using a Hammerstein model," *IEEE Trans. Autom. Control*, vol. 11, no. 3, pp. 546–550, 1966.
- [22] E.-W. Bai, "An optimal two-stage identification algorithm for Hammerstein–Wiener nonlinear systems," *Automatica*, vol. 34, no. 3, pp. 333–338, 1998.
- [23] R. Sameni, M. B. Shamsollahi, C. Jutten, and G. D. Clifford, "A nonlinear Bayesian filtering framework for ECG denoising," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 12, pp. 2172–2185, 2007.
- [24] H.-D. Park and O. Blanke, "Heartbeat-evoked cortical responses: mechanisms, functional roles, and methodological considerations," *NeuroImage*, vol. 197, pp. 502–511, 2019.